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Class : L.Y. B.Tech Batch : B3	Subject :
DOP :	DOC :

Practical No. – 13

Aim : Generate ANDNOT function using McCulloch-Pitts neural net by a MATLAB program

Theory :

McCullogh-Pitts Model

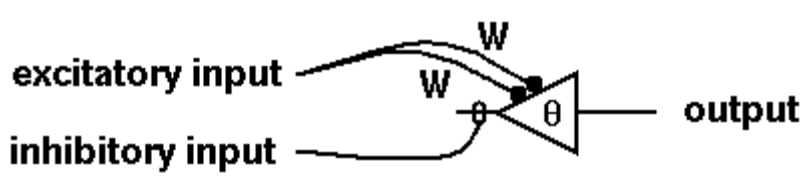
In 1943 two electrical engineers, Warren McCullogh and Walter Pitts, published the first paper describing what we would call a neural network. Their "neurons" operated under the following assumptions:

1. They are binary devices (V_i = [0,1])
2. Each neuron has a fixed threshold, theta
3. The neuron receives inputs from excitatory synapses, all having identical weights. (However it my receive multiple inputs from the same source, so the excitatory weights are effectively positive integers.)
4. Inhibitory inputs have an absolute veto power over any excitatory inputs.
5. At each time step the neurons are simultaneously (synchronously) updated by summing the weighted excitatory inputs and setting the output (V_i) to 1 iff the sum is greater than or equal to the threhold AND if the neuron receives no inhibitory input.

We can summarize these rules with the McCullough-Pitts output rule

$$V_i = \begin{cases} 1 & : \quad \sum_j W V_j \geq \theta \text{ AND no inhibition} \\ 0 & : \quad \text{otherwise} \end{cases}$$

and the diagram

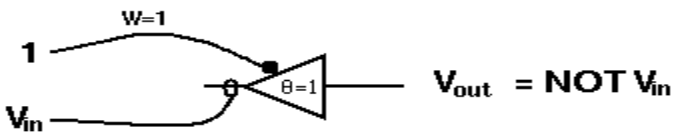


Using this scheme we can figure out how to implement any Boolean logic function. As you probably know, with a NOT function and either an OR or an AND, you can build up XOR's, adders, shift registers, and anything you need to perform computation.

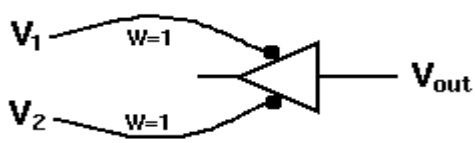
We represent the output for various inputs as a **truth table**, where 0 = FALSE, and 1 = TRUE. You should verify that when W = 1 and theta = 1, we get the truth table for the logical NOT,

```
Vin   |  Vout
-----+------
  1   |   0
  0   |   1
```

by using this circuit:



With two excitatory inputs V_1 and V_2, and W =1, we can get either an OR or an AND, depending on the value of theta:



if $\theta = 1 \Longrightarrow V_{out} = V_1 \text{ OR } V_2$

if $\theta = 2 \Longrightarrow V_{out} = V_1 \text{ AND } V_2$

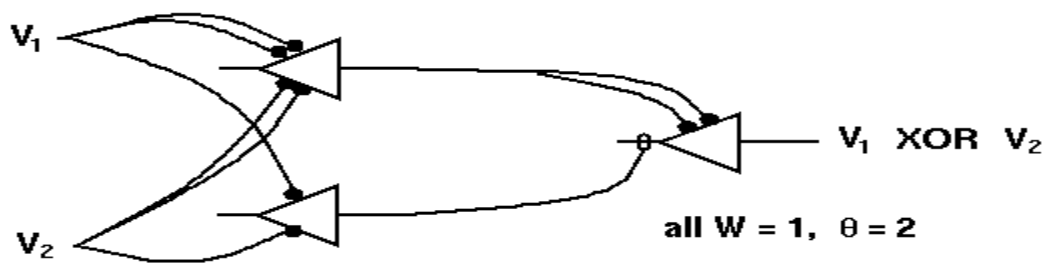
Can you verify that with these weights and thresholds, the various possible inputs for V_1 and V_2 result in this table?

```
V1 | V2 | OR | AND
---+----+----+----
 0 |  0 |  0 |  0
 0 |  1 |  1 |  0
 1 |  0 |  1 |  0
 1 |  1 |  1 |  1
```

The **exclusive OR** (XOR) has the truth table:

```
V1  | V2  | XOR
---+----+----
 0 |  0 |  0
 0 |  1 |  1        (Note that this is also a
 1 |  0 |  1         "1 bit adder".)
 1 |  1 |  0
```

It cannot be represented with a single neuron, but the relationship
XOR = (V_1 OR V_2) AND NOT (V_1 AND V_2) suggests that it can be represented with the network



Code :

```
disp('Enter weights');
w1=input('Weight w1=');
w2=input('weight w2=');
disp('Enter Threshold Value');
theta=input('theta=');
y=[0 0 0 0];
x1=[0 0 1 1];
x2=[0 1 0 1];
z=[0 0 1 0];
con=1;
while con
zin=x1*w1+x2*w2;
for i=1:4
if zin(i)>=theta
y(i)=1;
else
y(i)=0;
end
end
disp('Output of Net');
disp(y);


if y==z
```

```
con=0;
else
disp('Net is not learning enter another set of weights and Threshold
value');
w1=input('weight w1=');
w2=input('weight w2=');
theta=input('theta=');
end
end
disp('Mcculloch-Pitts Net for ANDNOT function');
disp('Weights of Neuron');
disp(w1);
disp(w2);
disp('Threshold value');
disp(theta);
```

Output :

```
Enter weights
Weight w1=
1
weight w2=
1
Enter Threshold Value
theta=
0.1
Output of Net
     0     1     1     1

Net is not learning enter another set of weights and Threshold value
weight w1=
1
weight w2=
1
theta=
1
Output of Net
     0     1     1     1

Net is not learning enter another set of weights and Threshold value
weight w1=
1
weight w2=
-1
theta=
1
Output of Net
     0     0     1     0

Mcculloch-Pitts Net for ANDNOT function
Weights of Neuron
     1

    -1

Threshold value
     1
```

Conclusion : In this practical we implemented ANDNOT function using McCulloch-Pitts neural modal in MATLAB.

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